

potential users, they found that people actually enjoyed time in their garden, so they reduced the scope of the device to simply monitor and inform the user when watering is required so that they could still attend to their plants on their own. However, when the user is not available, the device may be programmed to automatically water the garden.

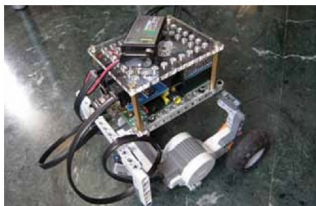
The open twist: H2O IQ is a completely open source device with all design information available online at <https://github.com/valkyriesavage/luffy-toboggans>. The 3D printed device is to be placed in the soil right next to the plant. At present, the prototype is programmed only for tomato plants, but its scope will be broadened. The device has a moisture sensor built with two galvanised nails submerged in plaster. A solar panel printed on the top charges the small battery that powers the device. H2O IQ also has an XBee radio on-board, which communicates moisture readings to a Raspberry Pi located at the edge of the garden. In order to conserve power, the microcontroller wakes up just once an hour for this transmission. When the device powers on, the Pi uses the opportunity to update the watering instructions if there is a new user request, and the ideal watering curve is programmed in it. The Pi also acts as a Web server, feeding information to a page with a Google Stocks powered display of historical moisture data and the ideal readings for the selected species of plant. Users can access this site online to view alerts, to set up auto watering, or to reprogram the device in other ways.

Turn your Raspberry Pi into a robot

<http://www.dexterindustries.com/BrickPi/>

Dexter Industries is a company that develops robotic sensors for LEGO Mindstorms NXT that are easy and fun to use, yet capable of being used in real-life applications. Its new project, BrickPi, is an open source slide-on board for the Raspberry Pi that helps you connect LEGO blocks, sensors, motors and parts to easily turn the compact Raspberry Pi into a powerful robot. You can connect up to three motors and up to four sensors, both digital and analogue.

The open twist: The BrickPi is based on the Atmel Atmega 328, and is powered by Arduino. Complete instructions on building, setting up and using the BrickPi are available online. The system consists of easily available components listed in the bill of materials, a circuit board, and a spot of laser cutting for which you might need to use an external service. The Eagle board and schematic are available on the Github repository (<https://github.com/DexterInd/BrickPi>), and so are the CAD files for the top and bottom parts of BrickPi's acrylic case. Assembling the BrickPi is relatively easy. While a little surface mount soldering is required, most of the parts are through-hole. Once you have all this set, you need to flash up the firmware. There is, however, no USB communication for the Arduino and it must be programmed using an ISP programmer, such



as the AVRISP mkII from Atmel. Finally, you need to prepare your BrickPi for operation by setting up I2C communications and Wi-Fi, and running the Python scripts. Then, you are all set to start using the robot! If you love it, you could even consider helping the team – its website reports that it needs some better Python code for controlling the BrickPi.

Knowing the air quality in your neighbourhood

<http://www.lvaqi.org/2013/>

Little Village Air Quality Initiative is an interesting open source project that aims to measure and visualise the air quality in an area. The project is still under development but the concept is interesting and definitely inspirational – worth helping or giving an independent try. The LVAQI uses multiple sensors programmed to collect data on the levels of carbon monoxide and nitrogen dioxide, temperature, humidity, etc, in real-time. Visualisations written in C++ using openFrameworks will be used to convert the data into graphical representations. There will be multiple Raspberry Pi computers with displays set up in public locations to display the air quality visualisations to help people understand the levels of pollution better.



The open twist: The board has been designed using Eagle CAD software and made with a CNC engraving machine. It is now ready for silk-screening and manufacture, and will be released as open hardware.